

Active Threat Hunting Lab Using Wazuh & Velociraptor

Built a home SOC lab to simulate endpoint threats and investigate malicious activity using **Wazuh**, **Velociraptor**, **Sysmon**, and **Microsoft Defender**. Correlated endpoint telemetry, Windows event logs, and security alerts to reconstruct the attack lifecycle.

Environment

Component	Technology
SIEM	Wazuh 4.x
DFIR Platform	Velociraptor 0.77
Endpoint Monitoring	Sysmon
Operating System	Windows 11 VM
Attacker Machine	Ubuntu Linux
Attack Framework	Metasploit Framework
Virtualization	VirtualBox

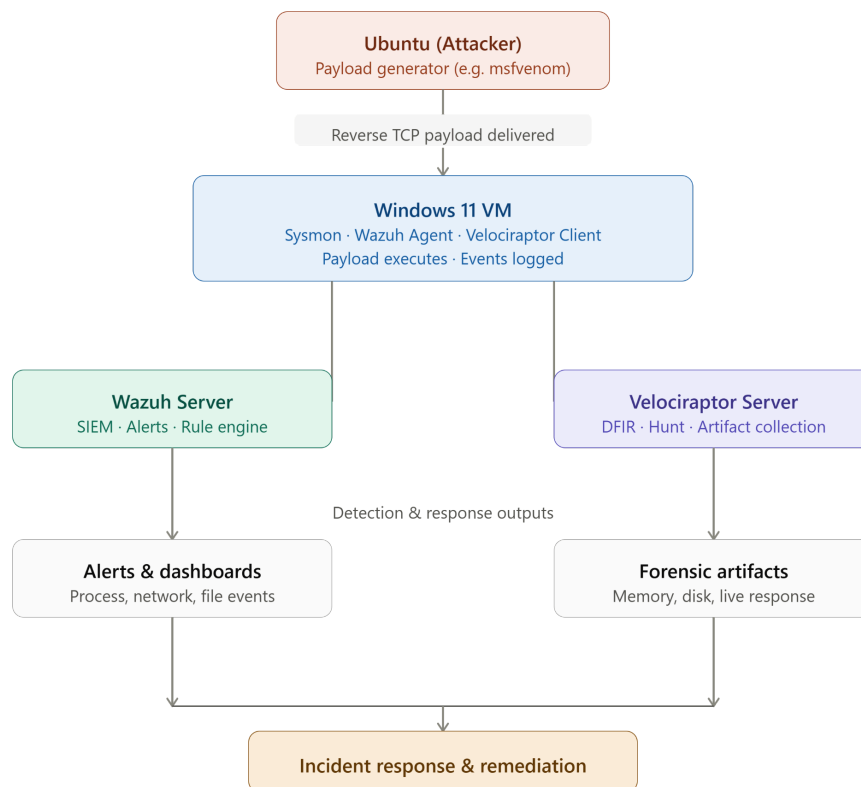
Tools Used

- Wazuh
- Velociraptor
- Sysmon
- Microsoft Defender
- Metasploit Framework
- Ubuntu Linux
- Windows 11
- PowerShell
- VirtualBox

Skills Demonstrated

- Security monitoring
- Threat hunting
- Endpoint detection and response (EDR)
- Windows event log analysis
- Sysmon analysis
- Digital forensics
- SIEM alert investigation
- Incident timeline reconstruction
- Threat correlation
- Malware execution analysis
- Network connection investigation

Lab Architecture



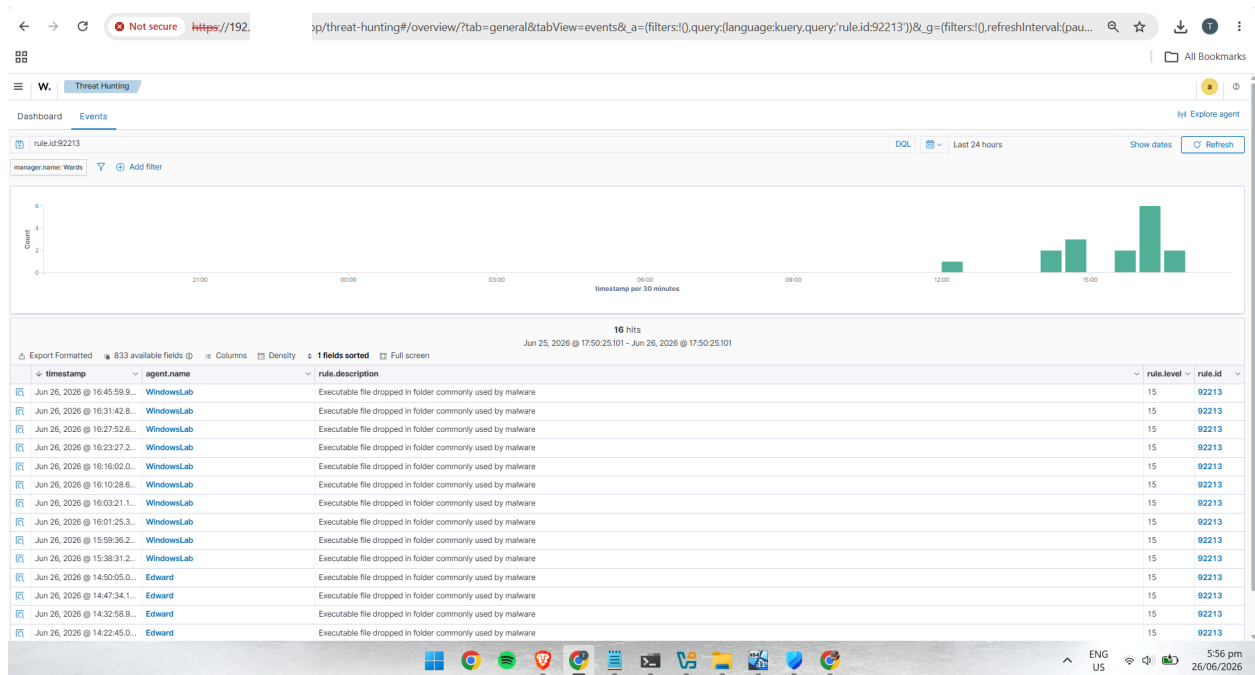
Attack Simulation

Performed a controlled attack simulation inside an isolated virtual lab.

Steps

- Generated a Meterpreter reverse TCP payload using Metasploit.
- Hosted the payload on the Ubuntu server.
- Downloaded the payload onto the Windows virtual machine using PowerShell.
- Executed the payload inside the isolated Windows VM.
- Established a reverse TCP connection to the Metasploit handler.
- Observed Windows Defender detecting and remediating the executable.

Investigation Using Wazuh



The screenshot displays the Wazuh Threat Hunting dashboard. On the left, a list of events is shown with columns for timestamp, agent name, and rule description. The events are filtered by rule ID 92213. On the right, the 'Document Details' panel shows the JSON data for the selected event. The event details include the agent name 'Wind', the event data image 'C:\WINDOWS\System32\WindowsPowerShell\v1.0\powershell.exe', the process ID '9968', the rule name 'EHE', the target file 'C:\Users\vboxuser\AppData\Local\Temp\test_shell.exe', the user 'WINDOWSLAB\vboxuser', the system channel 'Microsoft-Windows-Sysmon/Operational', the system computer 'WindowsLab', the event ID '11', the system event record ID '1235', the system keywords '0x8000000000000000', the system level '4', the system message 'File created: RuleName: EHE, UtcTime: 2026-06-26 08:31:39.315, ProcessGuid: {20b796e9-33f1-6a3e-8585-000000000200}, Image: C:\WINDOWS\System32\WindowsPowerShell\v1.0\powershell.exe, TargetFileName: C:\Users\vboxuser\AppData\Local\Temp\test_shell.exe', the system opcode '0', the system process ID '11212', and the system provider GUID '{0778385f-c22a-43eb-bf4c-b6f5698fbd9}'.

timestamp	agent name	rule description
Jun 26, 2026 @ 16:45:59.6...	WindowsLab	Executable file dropped in folder commonly used by malware
Jun 26, 2026 @ 16:31:42.8...	WindowsLab	Executable file dropped in folder commonly used by malware
Jun 26, 2026 @ 16:27:52.6...	WindowsLab	Executable file dropped in folder commonly used by malware
Jun 26, 2026 @ 16:23:27.2...	WindowsLab	Executable file dropped in folder commonly used by malware
Jun 26, 2026 @ 16:18:02.8...	WindowsLab	Executable file dropped in folder commonly used by malware
Jun 26, 2026 @ 16:10:26.6...	WindowsLab	Executable file dropped in folder commonly used by malware
Jun 26, 2026 @ 16:03:21.1...	WindowsLab	Executable file dropped in folder commonly used by malware
Jun 26, 2026 @ 16:01:25.3...	WindowsLab	Executable file dropped in folder commonly used by malware
Jun 26, 2026 @ 15:59:36.2...	WindowsLab	Executable file dropped in folder commonly used by malware
Jun 26, 2026 @ 15:36:31.2...	WindowsLab	Executable file dropped in folder commonly used by malware
Jun 26, 2026 @ 14:50:05.0...	Edward	Executable file dropped in folder commonly used by malware
Jun 26, 2026 @ 14:47:34.1...	Edward	Executable file dropped in folder commonly used by malware
Jun 26, 2026 @ 14:32:58.9...	Edward	Executable file dropped in folder commonly used by malware
Jun 26, 2026 @ 14:22:45.0...	Edward	Executable file dropped in folder commonly used by malware

Detection

Rule ID : 92213

Severity : Level 15 (High)

Alert : Executable file dropped in folder commonly used by malware

Evidence Collected

- Sysmon Event ID 11 (File Create)
- PowerShell created `test_shell.exe`
- Executable written to:

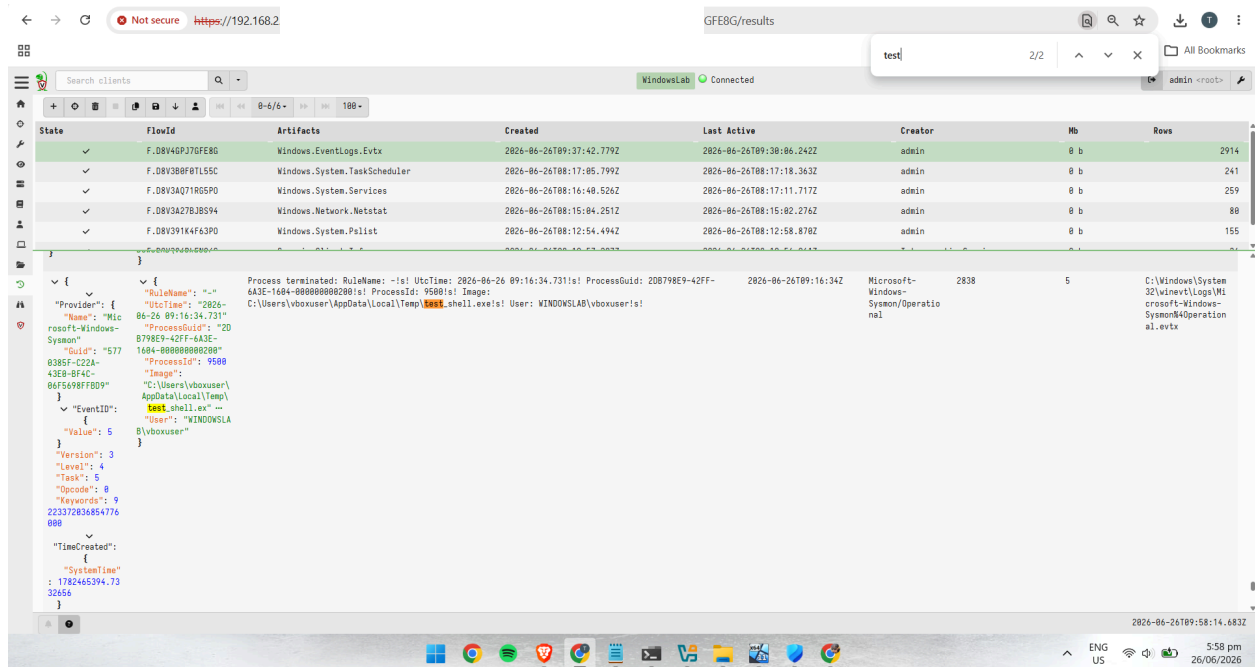
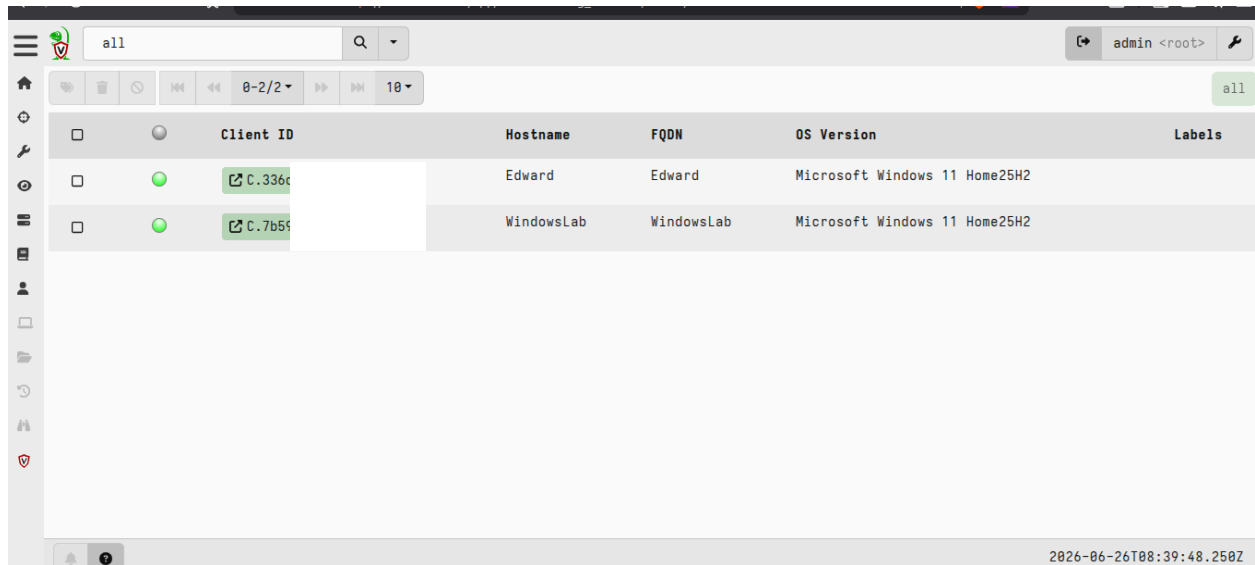
C:\Users\vboxuser\AppData\Local\Temp\

Outcome

Successfully identified suspicious executable creation through Wazuh correlation rules.

Investigation Using Velociraptor

Collected Windows forensic artifacts directly from the endpoint.



Artifacts Used

Windows.EventLogs.Evtx

Windows.System.Pslist

Windows.System.Netstat

Evidence Found

Successfully located Sysmon events including:

- Event ID 1 – Process Create
- Event ID 3 – Network Connection
- Event ID 5 – Process Termination
- Event ID 11 – File Creation

Observed:

- `test_shell.exe`
- `powershell.exe`
- Process execution timeline
- Event log artifacts from the Windows endpoint

[illegible]

```

ERROR: [Errno 98] Address already in use
wrds@Wards:~$ cd ~
python3 -m http.server 8080
Serving HTTP on 0.0.0.0 port 8080 (http://0.0.0.0:8080/) ...
192.168.2.100 - - [26/Jun/2026 16:23:26] "GET /test_shell.exe HTTP/1.1" 200 -
192.168.2.100 - - [26/Jun/2026 16:27:51] "GET /test_shell.exe HTTP/1.1" 200 -
192.168.2.100 - - [26/Jun/2026 16:31:41] "GET /test_shell.exe HTTP/1.1" 200 -

```

Correlation Timeline

Time	Event
Payload downloaded	PowerShell created <code>test_shell.exe</code>
Sysmon Event 11	File created
Wazuh	High severity alert generated
Sysmon Event 1	Payload executed
Sysmon Event 3	Reverse TCP connection initiated
Metasploit	Meterpreter session established
Sysmon Event 5	Process terminated
Microsoft Defender	Payload quarantined

File Hash Analysis

After downloading and executing the simulated Meterpreter payload (`test_shell.exe`), a cryptographic hash was generated using the Windows PowerShell `Get-FileHash` cmdlet. This hash serves as a unique identifier for the executable and can be used as an Indicator of Compromise (IOC) during threat hunting and malware analysis.

Hash Details

Attribute	Value
File Name	test_shell.exe
Hash Algorithm	SHA-256
Hash Value	02AA447EED4D604771CDAE539443FC0AC6481EB9B765FC8151... (truncated)
Purpose	Unique file identifier for malware analysis and IOC documentation

The generated SHA-256 hash serves as an **Indicator of Compromise (IOC)** and can be searched in malware intelligence platforms such as VirusTotal to determine whether the file has been previously identified as malicious. During this lab, the hash was documented together with evidence collected from Wazuh, Velociraptor, Sysmon, and Microsoft Defender to support the overall incident investigation.